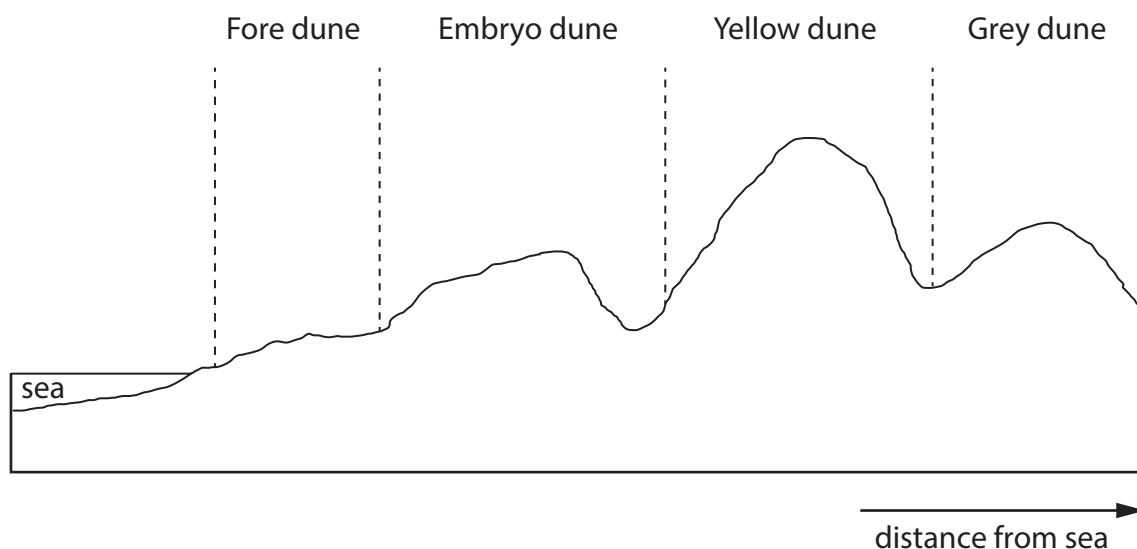


7 Stages of succession can be seen on different parts of a sand dune system.

The greater the distance of the sand dune from the sea, the later the stage of succession.

The diagram shows a sand dune system and the table gives some information about the different types of sand dune in this system.



Information	Fore dune	Embryo dune	Yellow dune	Grey dune
Soil depth / cm	<0.5	0.5	1.0	8.0
Percentage of organic matter (%)	<1.0	1.0	2.5	5.0
pH	8.5	8.0	7.0	6.5
Percentage of bare ground (%)	>97	97	70	10
Number of different plant species	2	3	6	15
Typical plant species	sea rocket saltwort	sea couch grass lyme grass marram grass	marram grass red fescue grass sea holly	range of different meadow plants

- (a) (i) The pH of a solution is a measure of the concentration of hydrogen ions in that solution.

It is a log scale e.g. a solution of pH 5 contains $10^{-5} \text{ mol dm}^{-3}$ of hydrogen ions.

What is the difference in concentration of hydrogen ions in a fore dune compared with a grey dune?

(1)

- ☐ **A** a fore dune has 2 times more hydrogen ions than the grey dune
- ☐ **B** a fore dune has 2 times fewer hydrogen ions than the grey dune
- ☐ **C** a fore dune has 100 times more hydrogen ions than the grey dune
- ☐ **D** a fore dune has 100 times fewer hydrogen ions than the grey dune

- (ii) Calculate the percentage increase in the number of plant species on the grey dune compared with the fore dune.

(1)

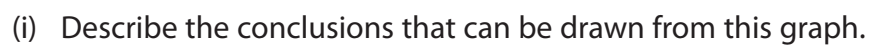
Answer.....%

*(iii) Explain the changes in these sand dunes with distance from the sea.

Use the information in the table to support your answer.

(6)

The graph shows the mean net primary productivity (NPP) above the ground and below the ground for these three types of sand dune.

[illegible]

(ii) Explain how the products of the light-independent reactions become NPP below the ground.

(4)

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(Total for Question 7 = 15 marks)
